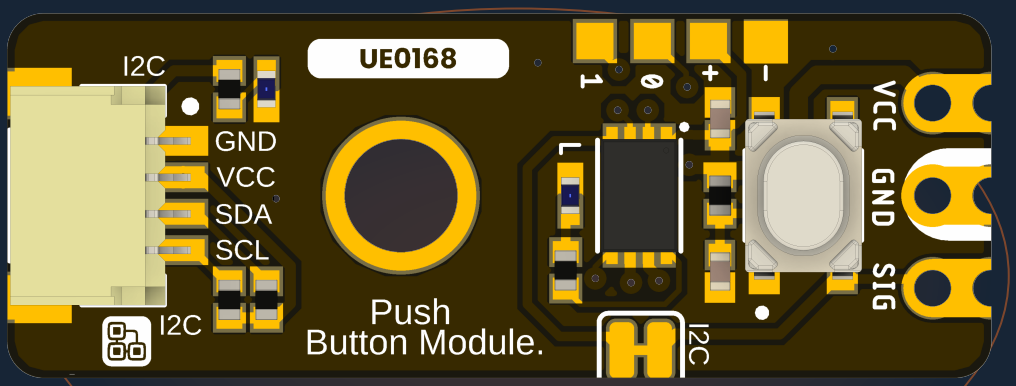
PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM

# DevLab I2C Push Button Module

**MPN: UE0168**

Version 1.1.0




I2C push-button module for digital user input.

PUSH BUTTON | I2C | DEVLAB ECOSYSTEM

DOCUMENT MAP

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# Description

The DevLab I2C Push Button Module (MPN **UE0168**) is an I2C push button input module in the UNIT Electronics DevLab family. The product-specific board views and any released drawings are collected under `hardware/`.

## Applications

- Push button input in an I2C-based prototype.
- Evaluation of the module with a compatible host controller.

## Available Documentation

A product-specific schematic was not present in the selected source package. See Chapter 8 for the available source files. Confirm the electrical interface against the product-specific pinout or schematic before connecting the board.

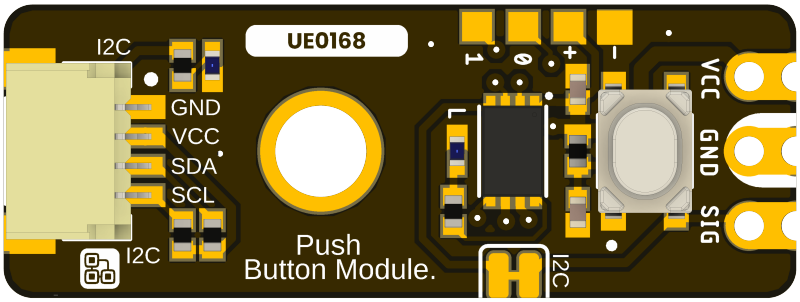
# 1 The Board

## 1.1 Board Identification

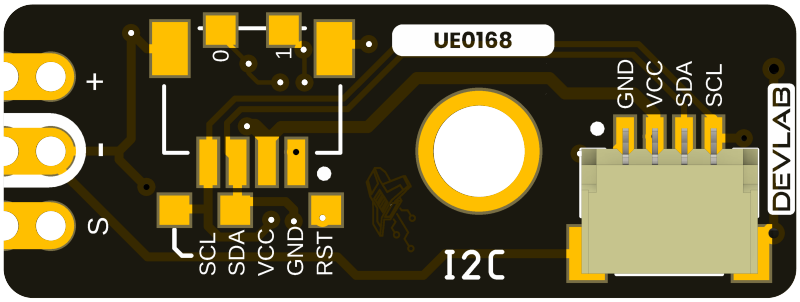
| Item | Identification |
|---|---|
| Product | DevLab I2C Push Button Module |
| Manufacturer Part Number (MPN) | UE0168 |
| Function | Push button input |

## 1.2 Board Views

The following views come from the selected UE0168 source package. They show component placement and board orientation; use a dimensioned drawing for mechanical tolerances.




*DevLab I2C Push Button Module top view.*




*DevLab I2C Push Button Module bottom view.*

# 2 Ratings

The selected UE0168 source package does not establish a complete set of electrical ratings in this reference. Do not use values from another DevLab module as substitutes.

## 2.1 Electrical Limits

Supply range, I/O voltage limits, and current consumption require confirmation from the product-specific schematic and component data before integration.

## 2.2 Interface Characteristics

Confirm the I2C address, bus speed, pull-up arrangement, and connector orientation with the released UE0168 pinout or firmware documentation. An I2C scan only identifies responding addresses; it does not verify the device protocol.

# 3 Functional Overview

The DevLab I2C Push Button Module provides push button input for an I2C-based system. The supplied board views identify the physical module, while the product-specific schematic and firmware documentation are needed to establish its internal signal path and command set.

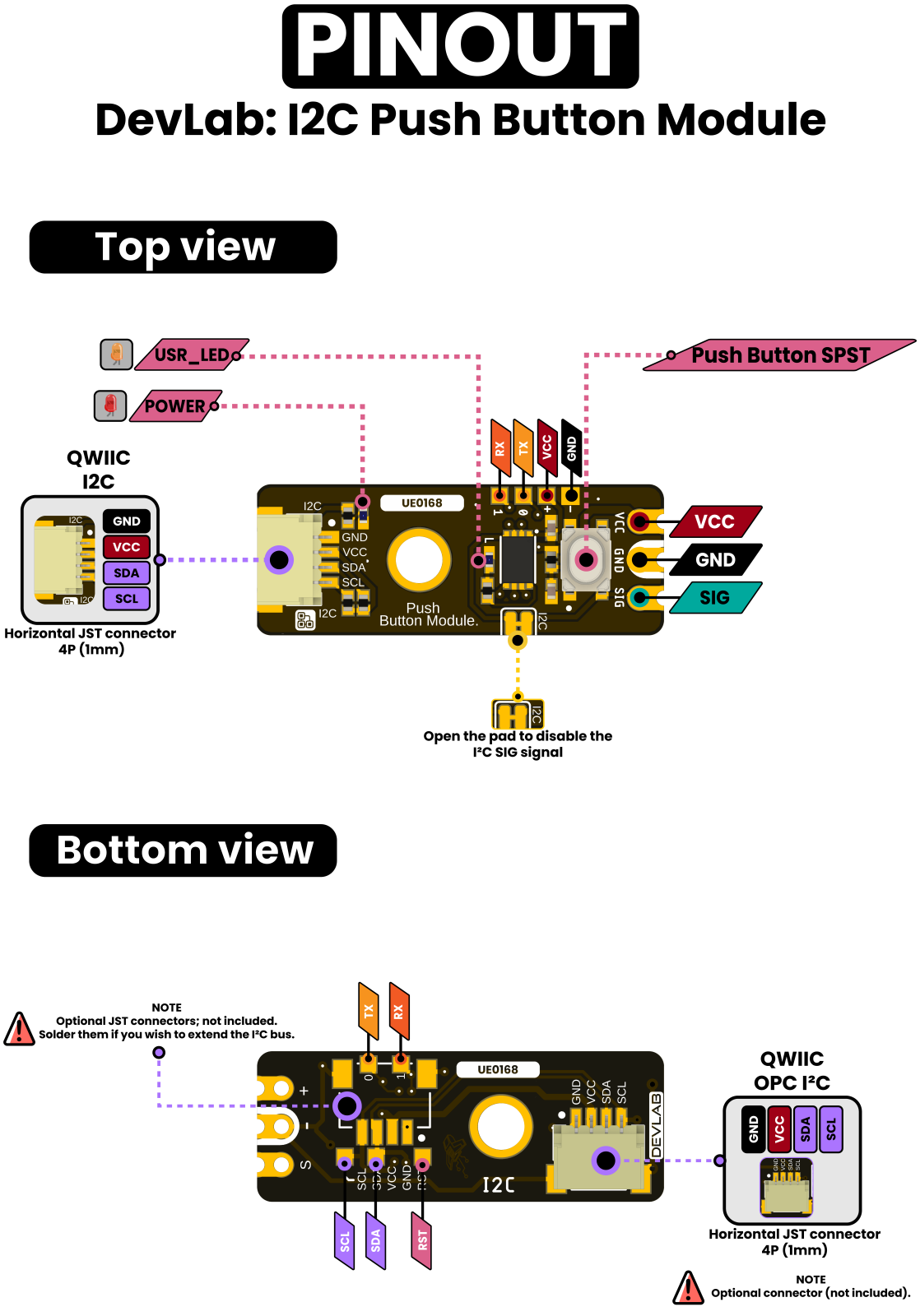
## 3.1 Integration Boundary

Use the available UE0168 pinout, schematic, and firmware release to determine power, connector mapping, address, and commands. Until these are confirmed, treat the generic I2C scanner in `software/` as a bus diagnostic only.

# 4 Connectors & Pinouts

The released V1.0 pinout for the UE0168 Push Button Module shows the top and bottom board views. Check the view and connector orientation before wiring.

## 4.1 Released Pinout




*UE0168 Push Button Module pinout, top and bottom views.*

**Figure 4.1 — UE0168 pinout, English version.** The Spanish version is linked in Section 8.

## 4.2 Connection Groups

| Group | Labels in the released drawing | Purpose |
|---|---|---|
| Qwiic I2C connector | GND, VCC, SDA, SCL | Power and I2C bus connections |
| Direct contacts | VCC, GND, SIG | Power and direct signal access |
| Service pads | TX, RX, RST | Pads identified in the drawing; consult the hardware design before use |

The drawing also marks two optional 4-pin, 1 mm JST connector positions. Its solder-link note says opening the marked pad disables the I2C SIG signal. The pinout does not establish electrical limits or connector part numbers.

# 5 Board Operation

## 5.1 Before Connecting

Check the product-specific UE0168 pinout or schematic for power, ground, SDA, and SCL. The board views alone do not establish voltage limits or connector numbering.

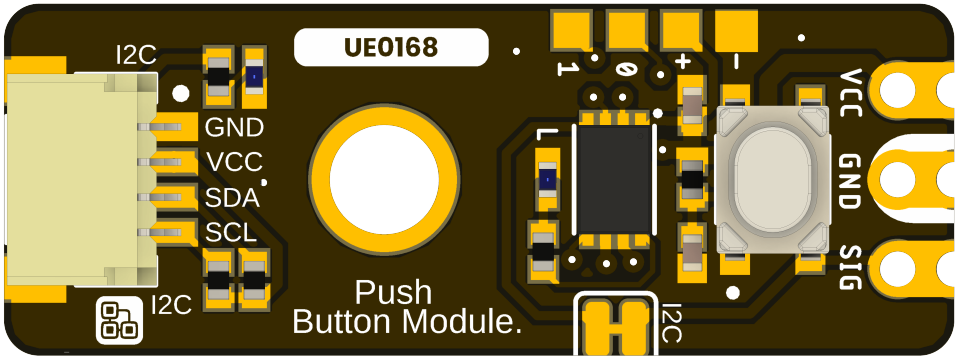
## 5.2 I2C Bus Check

After confirming the wiring and supply limits, use the generic Arduino or MicroPython I2C scanner in `software/examples/i2c/`. Set the host SDA and SCL pins for your controller. A responding address confirms only that a device acknowledged on the bus; it does not identify the module or validate its measurement protocol.

# 6 Mechanical Information

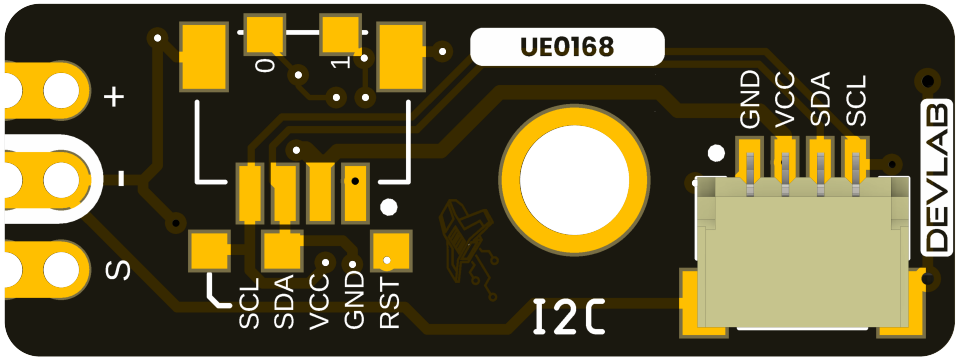
The supplied board views identify the component and underside layouts of the UE0168 I2C Push Button Module. Use the labeled drawings for fit checks; do not infer dimensions by scaling a render.

## 6.1 Top View



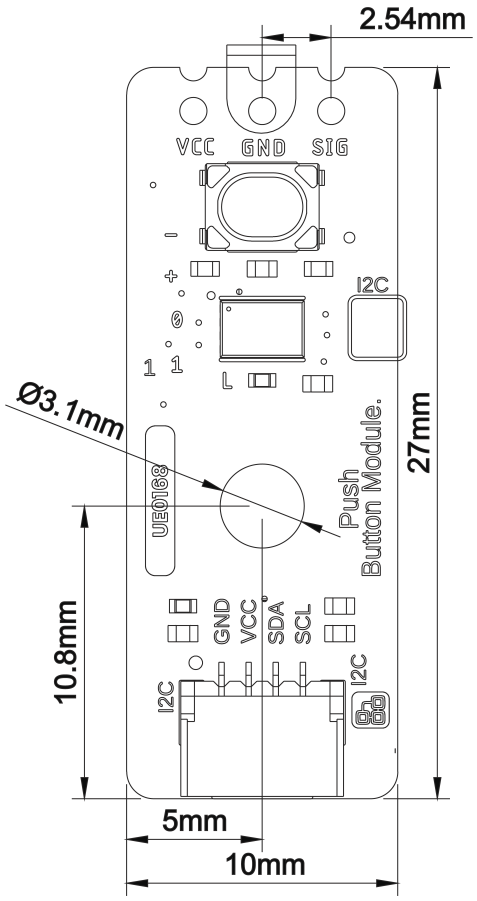

*UE0168 I2C Push Button Module top view.*

## 6.2 Bottom View




*UE0168 I2C Push Button Module bottom view.*

## 6.3 Board Dimensions




**Board Dimensions**

*UE0168 I2C Push Button Module dimension drawing.*

Use the dimensions printed on the supplied drawing. PCB thickness, component heights, connector clearance, and manufacturing tolerances require separate verification.

# 7 Company Information

| Item | Information |
|---|---|
| Company / brand | UNIT Electronics |
| Board ecosystem | DevLab |
| Product family | Atom |
| Website | UNIT Electronics |
| Address | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico |

The product repository and controlled technical references are listed in the next chapter.

# 8 Reference Documentation

| Reference | Location |
|---|---|
| Product repository | UE0168 Push Button Module |
| Pinout V1.0, English | PDF · PNG |
| Pinout V1.0, Spanish | PDF · PNG |
| Top view | 3D board view |
| Bottom view | 3D board view |
| Board dimensions | Dimension drawing |
| Board topology | Topology drawing |
| CAD model | STEP |

The released pinout documents the visible labels and connector orientation. It does not by itself establish supply limits, bus timing, or mating part numbers.

# 9 Appendix

## 9.1 Schematic Status

A product-specific schematic was not included in the selected source package.

## 9.2 Document Control

| Item | Value |
|---|---|
| Product | DevLab I2C Push Button Module |
| MPN | UE0168 |
| Source | UNIT Electronics selected product source package |

## 9.3 Data Requiring Confirmation

Electrical limits, I2C commands, and mechanical tolerances must be verified against product-specific released documentation.